Web Programming Lab 7 - Recitation

Question 1

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React Portfolio Components</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
    <style>
        * { margin: 0; padding: 0; box-sizing: border-box; }
        body { font-family: 'Segoe UI', sans-serif; background: #f5f7fa; }
        .header { background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%); color: white; padding: 60px 20px
        .header h1 { font-size: 3em; margin-bottom: 10px; }
        .header p { font-size: 1.3em; opacity: 0.95; }
        .section { max-width: 1000px; margin: 40px auto; padding: 35px; background: white; border-radius: 15px; b
        .section h2 { color: #f5576c; border-bottom: 3px solid #f093fb; padding-bottom: 12px; margin-bottom: 25px
        .section p { line-height: 1.8; color: #555; font-size: 1.1em; }
        .skills-grid { display: flex; gap: 15px; flex-wrap: wrap; margin-top: 10px; }
        .skill-badge { background: linear-gradient(135deg, #f093fb 0%, #f5576c 100%); color: white; padding: 12px
        .skill-badge:hover { transform: translateY(-3px); box-shadow: 0 5px 15px rgba(245, 87, 108, 0.4); }
        .project-card { border: 2px solid #f0f2f5; border-radius: 12px; padding: 25px; margin: 15px 0; transition
        .project-card:hover { border-color: #f093fb; transform: translateY(-5px); box-shadow: 0 8px 25px rgba(240
        .project-card h3 { color: #2c3e50; font-size: 1.5em; margin-bottom: 10px; }
        .project-card p { color: #666; margin: 10px 0; line-height: 1.6; }
        .project-card small { color: #f5576c; font-weight: 600; background: #ffe5ec; padding: 5px 12px; border-ra
        .footer { text-align: center; padding: 30px; background: #2c3e50; color: white; margin-top: 40px; font-si
    </style>
</head>
<body>
    <div id="root"></div>
    <script type="text/babel">
        // Functional Component: Header
        function Header({ name, title }) {
            return (
                <header className="header">
                    <h1>■■■ {name}</h1>
                    <p>{title}</p>
                </header>
            );
        }

        // Functional Component: About
        function About() {
            return (
                <section className="section">
                    <h2>■ About Me</h2>
                    <p>I am a second-year AIML engineering student at SFIT, passionate about machine learning, wel
                </section>
            );

        }

        // Functional Component: Skills
        function Skills({ skills }) {
            return (
                <section className="section">
                    <h2>■ Skills</h2>
                    <div className="skills-grid">
                        {skills.map((skill, i) => <span key={i} className="skill-badge">{skill}</span>)}
                    </div>
                </section>
            );
        }

        // Class Component: Projects
        class Projects extends React.Component {
            render() {
                return (
                    <section className="section">
```

```
                    <h2>■ Projects</h2>
                    {this.props.projects.map((proj, i) => (
                        <div key={i} className="project-card">
                            <h3>{proj.name}</h3>
                            <p>{proj.description}</p>
                            <small>■ Tech: {proj.tech}</small>
                        </div>
                    ))}
                </section>
            );
        }
    }

    // Functional Component: Footer
    function Footer() {
        return <footer className="footer"><p>© 2026 Tony Rodrigues | Built with React ■■</p></footer>;
    }

    // Main App
    function App() {
        const skills = ['Python', 'Machine Learning', 'React', 'HTML/CSS', 'JavaScript', 'SQL', 'NumPy', 'Pan
        const projects = [
            { name: 'Disease Predictor', description: 'ML-based disease prediction system using Naive Bayes a
            { name: 'Portfolio Website', description: 'Responsive personal portfolio built with React compone
            { name: 'Data Pipeline', description: 'End-to-end ML pipeline with data preprocessing, feature en
        ];

        return (
            <div>
                <Header name="Tony Rodrigues" title="SE AIML Student | Web Developer | ML Enthusiast" />
                <About />

                <Skills skills={skills} />
                <Projects projects={projects} />
                <Footer />
            </div>
        );
    }

    ReactDOM.createRoot(document.getElementById('root')).render(<App />);
  </script>
</body>
</html>
```

Output:

👨‍🎓 Tony Rodrigues

SE AIML Student | Web Developer | ML Enthusiast

📖 About Me

I am a second-year AIML engineering student at SFIT, passionate about machine learning, web development, and building intelligent systems. My journey combines academic excellence with hands-on projects that solve real-world problems.

🚀 Skills

Python | Machine Learning | React | HTML/CSS | JavaScript | SQL | NumPy | Pandas | TensorFlow | Git

💼 Projects

Disease Predictor

ML-based disease prediction system using Naive Bayes and Neural Networks with 92% accuracy.

Tech: Python, Scikit-learn, Flask

Portfolio Website

Responsive personal portfolio built with React components and modern design principles.

Tech: React, HTML, CSS, JavaScript

Data Pipeline

End-to-end ML pipeline with data preprocessing, feature engineering, and model evaluation.

Tech: Python, Pandas, Sklearn, Matplotlib

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Question 2

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React Props - Dynamic Portfolio</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
    <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
    <style>
        * { margin: 0; padding: 0; box-sizing: border-box; }
        body { font-family: 'Segoe UI', sans-serif; background: linear-gradient(135deg, #ffecd2 0%, #fcb69f 100%)
        .header { background: linear-gradient(135deg, #FF6B6B 0%, #FFE66D 100%); color: white; padding: 60px 20px
        .header h1 { font-size: 3em; margin-bottom: 10px; }
        .header p { font-size: 1.3em; opacity: 0.95; }
        .section { max-width: 1000px; margin: 40px auto; padding: 35px; background: white; border-radius: 15px; b
        .section h2 { color: #FF6B6B; border-bottom: 3px solid #FFE66D; padding-bottom: 12px; margin-bottom: 25px
        .section p { line-height: 1.8; color: #555; font-size: 1.1em; }
        .profile-card { display: flex; gap: 25px; align-items: center; padding: 25px; background: linear-gradient
        .profile-card:hover { transform: translateY(-5px); box-shadow: 0 10px 30px rgba(255, 107, 107, 0.15); bor
        .profile-card .avatar { width: 90px; height: 90px; border-radius: 50%; background: linear-gradient(135deg
        .profile-card .info h3 { color: #2c3e50; margin-bottom: 8px; font-size: 1.4em; }
        .profile-card .info p { color: #666; font-size: 1.05em; }
        .footer { text-align: center; padding: 30px; background: #2c3e50; color: white; margin-top: 40px; font-si
    </style>
</head>
<body>
    <div id="root"></div>
    <script type="text/babel">
        // Functional Component with Props: Profile
        function Profile({ name, role }) {
            return (
                <header className="header">
                    <h1>■ {name}</h1>
                    <p>{role}</p>
                </header>
            );
        }

        // Class Component with Props: About
        class About extends React.Component {
            render() {
                return (
                    <section className="section">
                        <h2>■ About Me</h2>
                        <p>{this.props.description}</p>
                    </section>
                );
            }
        }

        // Functional Component with Props: TeamMember
        function TeamMember({ name, role, initial }) {
            return (
                <div className="profile-card">
                    <div className="avatar">{initial}</div>
                    <div className="info">
                        <h3>{name}</h3>
                        <p>■ {role}</p>
                    </div>
                </div>
            );
        }

        // Team Section using Props
        function Team({ members }) {
            return (
                <section className="section">
                    <h2>■ Team</h2>
                    {members.map((m, i) => (
                        <TeamMember key={i} name={m.name} role={m.role} initial={m.name[0]} />
                    ))}
                </section>
            );
        }

        function Footer({ year }) {
            return <footer className="footer"><p>© {year} Tony Rodrigues | Props Demo with React ■■</p></footer>;
```

```
        }

        // App: Passing Props from Parent to Children
        function App() {
            const teamMembers = [
                { name: "Tony Rodrigues", role: "Lead Developer & ML Engineer" },
                { name: "Amit Kumar", role: "UI/UX Designer" },
                { name: "Sarah Chen", role: "Backend Developer" },
                { name: "Mike Johnson", role: "Data Scientist" }
            ];

            return (
                <div>
                    <Profile name="Tony Rodrigues" role="SE AIML Student | Web Developer | React Enthusiast" />
                    <About description="I am a second-year AIML engineering student at SFIT, passionate about mach
                    <Team members={teamMembers} />
                    <Footer year={2026} />
                </div>
            );
        }

        ReactDOM.createRoot(document.getElementById('root')).render(<App />);

    </script>
</body>
</html>
```

Output:

👤 Tony Rodrigues

SE AIML Student | Web Developer | React Enthusiast

📖 About Me

I am a second-year AIML engineering student at SFIT, passionate about machine learning and web development. This page demonstrates passing data between React components using props - a core concept in React that enables component communication and reusability.

👥 Team

T **Tony Rodrigues**
Lead Developer & ML Engineer

A **Amit Kumar**
UI/UX Designer

S **Sarah Chen**
Backend Developer

M **Mike Johnson**
Data Scientist

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